Digestion & enzymes

Food → molecules

1 The digestive route

Mouth: chewing/amylase. Oesophagus: moves food. Stomach: acid, protease and mixing.

Pancreas supplies enzymes; small intestine digests/absorbs; large intestine absorbs water.

Egestion: undigested food out. **Excretion:** metabolic waste out.

2 Enzyme products

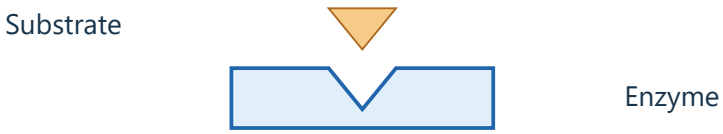
Amylase: **starch → sugars;** made in salivary glands, pancreas and small intestine.

Protease: **protein → amino acids;** stomach, pancreas, small intestine.

Lipase: **lipid → glycerol + fatty acids;** pancreas and small intestine.

How enzymes work

3 Specific active sites



Enzymes are reusable **protein catalysts.** Substrate binds a complementary active site.

Warming increases collisions to an optimum; excess heat/extreme pH can **denature** the active site. Cold slows activity.

4 Bile & absorbed products

Liver makes bile; gall bladder stores it. **Alkaline** bile neutralises stomach acid.

Emulsification makes fat droplets → more surface area for lipase. Bile is not an enzyme.

Products build new molecules; glucose can be respired.

Investigate

5 Food tests

Benedict's + water bath: reducing sugar → coloured precipitate. Iodine: starch → blue-black.

Biuret: protein → purple. Sudan III: lipid → red oil layer.

Separate samples; positive/negative controls; goggles and safe heating.

6 Amylase & pH practical

Vary buffer pH; control temperature, concentrations and volumes.

Sample onto fresh iodine every **30 s.** Endpoint: iodine stays orange-brown. Repeat.

Same starch amount: **relative rate = 1/time.** Shorter intervals improve endpoint resolution.